

Lebanon Mobility Intelligence: Discovering Recurring Road-Segment Bottlenecks from Crowdsourced Traffic Data

LebNet Tech Fellows — AI for Lebanon Project

1. Project Title & Abstract

Lebanon has significant mobility and congestion challenges, but publicly available, geolocated, longitudinal traffic data is scarce. This project uses the Tari'ak crowdsourced traffic dataset, 6,006,401 GPS observations spanning 1,673 dates and 14,289 mapped OpenStreetMap road segments, to identify recurring mobility bottlenecks: road segments and time windows that repeatedly show slower-than-usual movement relative to their own historical baseline. The pipeline moves from feasibility assessment and SQL-based feature engineering to unsupervised K-Means clustering of 559 sufficiently-observed segments (roughly 3.9% of the mapped network) into five behavioral archetypes, and concludes with a ranked candidate-bottleneck list and a geographic visualization. The project does not predict traffic, measure live congestion, or establish causality; it produces a historical, evidence-based screening layer intended to help transportation planners prioritize where further investigation is warranted.

2. Introduction & Problem Statement

Congestion and poor transport connectivity affect commuting time, access to services, and productivity across Lebanon, yet detailed, geolocated traffic data suitable for research or planning is limited. This project asks: can historical, crowdsourced movement observations be transformed into useful evidence about where and when recurring mobility problems appear? The intended users are municipalities and transportation planners, who could use the results to prioritize road segments for further investigation, targeted data collection, or infrastructure review, with public-transport planners, logistics operators, and mobility researchers as secondary beneficiaries. The project is deliberately scoped as a historical, decision-support prototype rather than a live traffic, accident-prediction, or congestion-causality system, since the dataset does not contain the traffic volume, road capacity, incident, or real-time information such systems would require.

3. Methodology

The project is organized into four notebooks representing successive stages of a standard data science workflow.

- **Feasibility (Notebook 00):** A full chunked scan of all 6,006,401 raw rows validated schema, timestamps, coordinates, and duplicates with no exact-duplicate or invalid records found. A targeted follow-up investigation characterized near-zero velocity readings (0.72% of rows, distributed broadly across hours and segments rather than concentrated in any single location) and confirmed they represent genuinely stationary GPS readings rather than a sensor artifact. These rows are retained in the cleaned dataset but excluded from velocity-based statistics to prevent small-sample averages, particularly at the segment-hour level, from being distorted by stopped-vehicle readings.
- **SQL Feature Engineering (Notebook 01):** SQLite is used to move from raw observations to a compact *segment_features* table: one row per road segment with median velocity, velocity variability, and AM (07:00–09:00) and PM (17:00–19:00) relative slowdown magnitude and frequency. Each segment's peak-hour baseline is computed strictly from its own non-peak hours, a correction made after an initial version diluted the slowdown signal by including peak hours in their own baseline. Segments are filtered to those with at least 100 total observations and 10 active days, a threshold chosen using an observation-count and active-days distribution histogram rather than asserted arbitrarily.

- **Unsupervised Clustering (Notebook 02):** Behavioral features are scaled with StandardScaler and passed to K-Means. Because AM/PM data is missing for many otherwise-qualified segments, the clustering cohort is further restricted to 559 segments with complete peak-hour coverage, a decision made to avoid imputing behavior for roads that were simply undersampled rather than genuinely free-flowing. Cluster count is selected using an elbow (inertia) and silhouette-score sweep across $k=2-10$; $k=5$ is chosen over the numerically higher-scoring $k=2$ because $k=2$ only separates segments into "fast" versus "slow" and discards the AM/PM temporal distinction central to the project's purpose. Cluster labels are assigned only after inspecting each cluster's feature distributions and the segment closest to each centroid, not decided in advance.
- **Visualization and Decision Support (Notebook 03):** A transparent candidate-severity score, relative slowdown multiplied by slowdown frequency, ranks segments by reliability of the pattern rather than by raw slowdown magnitude alone, so a single anomalous day cannot outrank a segment that is consistently slow. This ranking, a 24-hour temporal profile by cluster, and a geographic scatter of all 559 qualified segments with the top 20 candidates highlighted make up the final decision-support outputs.

4. Implementation Details & Results

The pipeline is implemented in Python (Pandas, NumPy, scikit-learn, Matplotlib) with SQLite as the analytical backend for the six-million-row dataset. Key quantitative results: 4,791 of 14,289 segments passed the observation-count and active-days threshold; 559 of those (~3.9% of the total mapped network) had complete AM/PM peak-hour coverage and formed the clustering cohort. This cohort has roughly 9 times the mean observation count and 2.7 times the mean active-days of the excluded segments, supporting the inference that it is weighted toward more heavily-traveled corridors. K-Means with $k=5$ achieved a silhouette score of 0.2097, indicating moderate cluster cohesion with real overlap between groups rather than sharp separation, consistent with traffic behavior existing on a continuous spectrum rather than in discrete categories. The five resulting archetypes, Severe Morning-Peak Bottlenecks, Severe Evening-Peak Bottlenecks, Dual-Peak Rush-Hour Sensitive, Steady Moderate-Speed Flow, and High-Speed Free-Flow Corridors, were independently corroborated by the Notebook 03 ranking: segments labeled Severe Morning-Peak consistently show an AM slowdown frequency near 1.0 alongside a negative or near-zero PM value, and Evening-Peak segments show the mirror pattern, agreement between two different analytical methods (distance-based clustering and frequency-weighted ranking) that were never designed to match. The final ranked candidate list and geographic map show the strongest candidates are not uniformly distributed: roughly two-thirds of the top 20 candidates cluster tightly in one geographic area consistent with the greater Beirut region, with a smaller number of outliers further south and near the Bekaa Valley.

5. Discussion & Analysis

The results support the project's central hypothesis: historical, crowdsourced velocity data does contain a recoverable, recurring signal at the segment and time-of-day level, and that signal can be organized into interpretable behavioral categories without a labeled ground truth. The cross-validation between clustering and ranking, two analytically distinct methods converging on the same segment characterizations, is a stronger result than either method alone would provide. That said, several limitations bound what can be concluded. The clustering cohort covers only about 3.9% of the mapped network; areas without qualifying segments reflect a lack of sufficient peak-hour observations, not verified free-flowing traffic, and this caveat is carried through every visualization in Notebook 03. The dataset contains no road-classification, road-capacity, incident, or travel-direction information, so while the geographic clustering of top candidates near Beirut is a real, visible pattern, this project cannot identify specific named corridors or attribute the pattern to a specific cause. The moderate silhouette score (0.21) means cluster boundaries are not sharp, and no stability testing across random seeds or bootstrap resampling was performed to confirm the archetypes are robust to resampling, a natural next step. Similarly, no temporal holdout was tested; whether "problem" corridors identified from this

multi-year window remain problematic in more recent data is untested and would substantially strengthen the project's practical claims if confirmed. Throughout, the project intentionally uses "candidate," "recurring," and "historical signal" rather than "confirmed congestion" or causal language, reflecting the genuine limits of what velocity, timestamp, and coordinate data alone can establish.

6. Reflection on Learnings

The most rewarding part of this project was closing gaps rather than avoiding them. An early investigation into near-zero velocity readings, initially assumed to be a simple data-quality issue, turned out to reveal that no exact-zero readings existed at all, only near-zero stationary readings, which changed how the exclusion was framed and documented. A more consequential discovery was a circularity bug in the AM/PM baseline calculation: an early version compared each segment's peak-hour velocity against a baseline that included the peak hours themselves, which systematically understated the very slowdowns the project was built to detect. Finding and correcting this, and verifying the fix propagated to both the slowdown-magnitude and slowdown-frequency features, was the single most technically significant moment of the project. Learning K-Means clustering from a standing start was the steepest personal learning curve: understanding the assign-and-update loop, why feature scaling is necessary before distance-based clustering, why a sigmoid transform would have been the wrong scaling choice for this dataset's velocity range, and what a silhouette score is actually measuring, point-to-point distance rather than distance to a centroid, took deliberate, incremental effort. The most challenging and clarifying moment was catching an unverified claim in a draft conclusion that attributed the geographic clustering of candidates to a named "coastal highway corridor," a plausible-sounding claim the dataset's coordinate-only geographic information could not actually support. Correcting it reinforced the project's central discipline: a finding is only as strong as the evidence directly behind it, and the temptation to state a more specific, more satisfying conclusion than the data supports is a real and recurring risk that requires active, continuous checking rather than one-time vigilance.